

Term 7- Sept 2025

Nanoelectronics and Technology
(01.119/99.503)-Week 5 Class 2

Shubhakar K
16-Oct- 2025

Email: shubhakar@sutd.edu.sg
Office location: 1.602-26

Outline

- Photolithography
- Double Patterning
- Etching
- E-beam lithography

Photolithography

- Lithography process refers to the formation of latent pattern on a polymer film (resist) coated on wafer surface which is then permanently transferred down to the underneath substrate through etching or implant process.

- **Patterning Layer Formation**

Done either by a growth or a deposition. This layer is to be selectively removed.

- **Photoresist Application**
- **Mask:**
- **Exposure to UV light**
- **Development and etching**

Photolithography

Photoresist

A **photoresist** is a **light-sensitive organic polymer** used to transfer a geometric pattern from a photomask onto a substrate (usually a silicon wafer) during the **photolithography** step of IC fabrication.

When exposed to **UV light**, **deep UV (DUV)**, or **EUV radiation**, the chemical structure of the photoresist changes, allowing **selective removal (development)** of either the exposed or unexposed areas.

Photoresist Composition

A typical photoresist formulation includes:

Component	Function
Polymer (Resin)	Forms the base film and provides mechanical stability
Photoactive Compound (PAC)	Sensitive to radiation; changes solubility upon exposure
Solvent	Controls viscosity; allows uniform spin coating
Additives	Improve adhesion, contrast, or thermal stability

Photolithography

- **Photoresist Application**

Feature	Positive Resist	Negative Resist
Exposure effect	Exposed region becomes soluble	Exposed region becomes insoluble
Pattern result	Mask = wafer pattern (direct)	Mask = inverse pattern (reversed)
Resolution	Higher (used for sub-100 nm features)	Lower (swelling, crosslinking blur)
Etch resistance	Lower	Higher
Applications	Mainstream IC lithography (g-line to EUV)	MEMS, packaging, thick resist, niche IC use

Key Applications

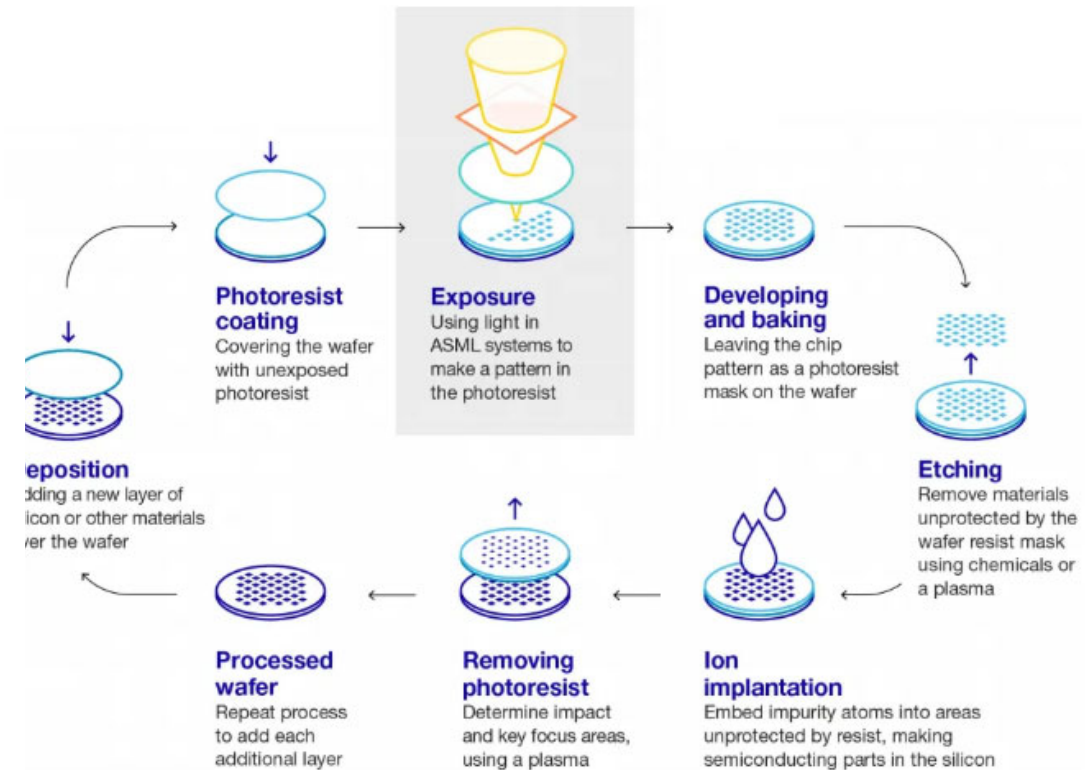
- Patterning active regions, gates, interconnects, vias

Photolithography

- Photoresist Application

Processing Steps Involving Photoresist

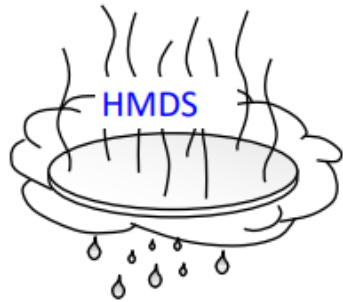
1. Surface Cleaning (remove contaminants)
2. HMDS Coating (promotes adhesion)
3. Spin Coating of Resist
4. Soft Bake (evaporate solvent)
5. Exposure (through photomask)
6. Post-Exposure Bake (in chemically amplified resists)
7. Development (using alkaline developer like TMAH)
8. Hard Bake (improve durability before etching)



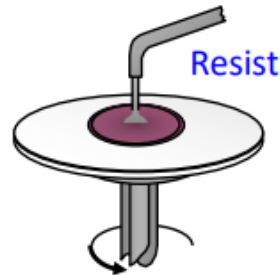
Photolithography

- Photoresist Application

HMDS (Hexamethyldisilazane) is used in photolithography as an adhesion promoter to ensure photoresist sticks properly to the substrate, typically a silicon wafer. This prevents issues like delamination and helps create sharper, more reliable patterns for semiconductor manufacturing.



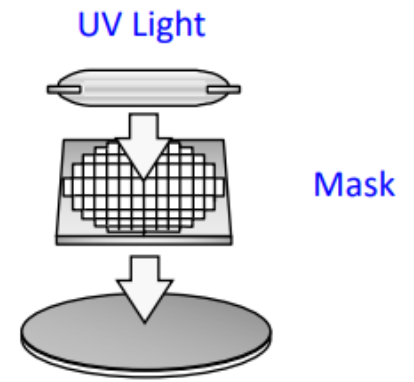
1) Vapor prime



2) Spin coat



3) Soft bake



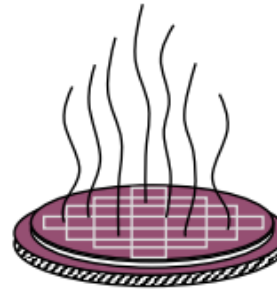
4) Alignment and exposure



5) Post-exposure bake



6) Develop

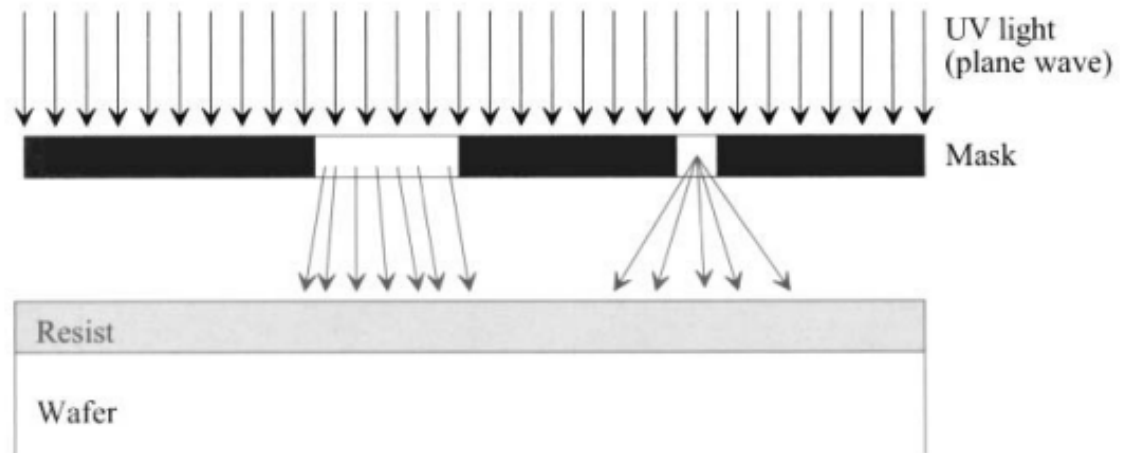
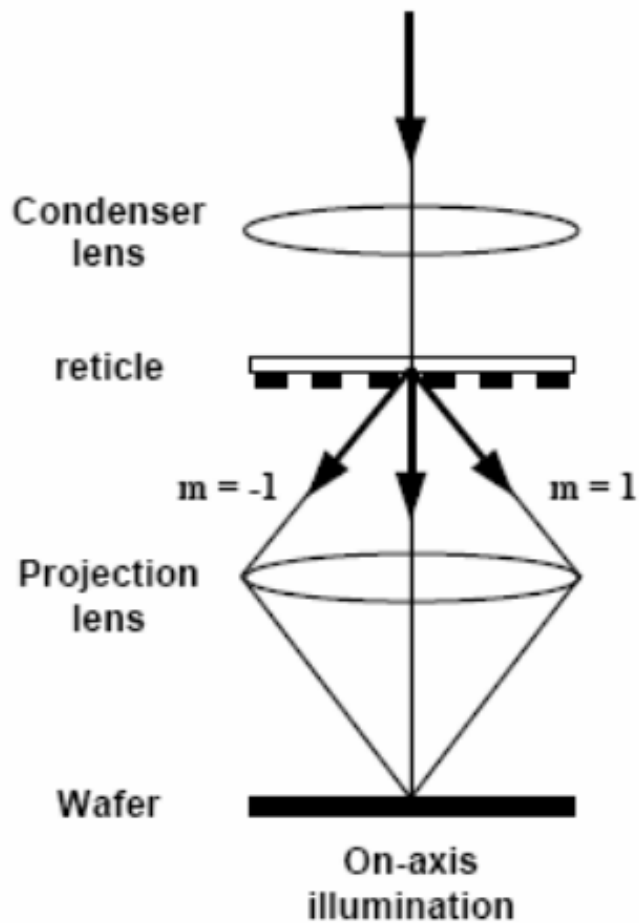


7) Post-develop bake

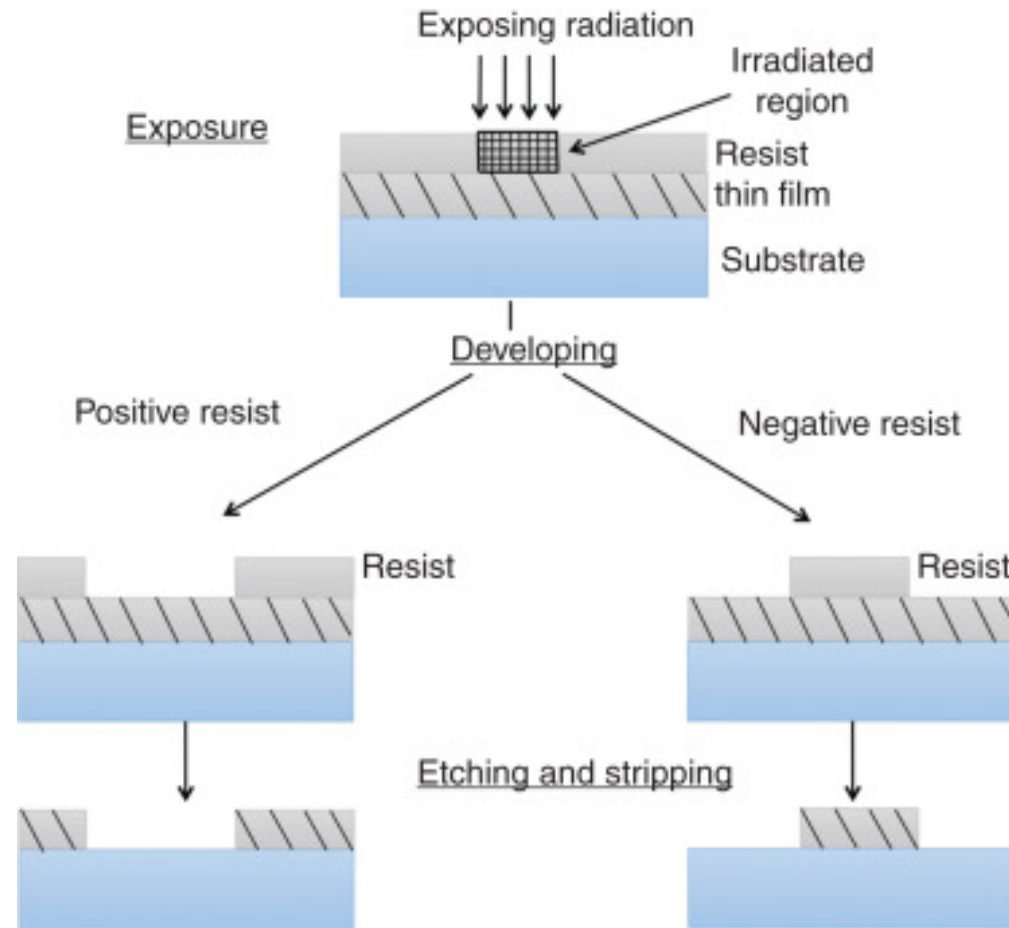


8) Optical inspection

Optical Imaging System

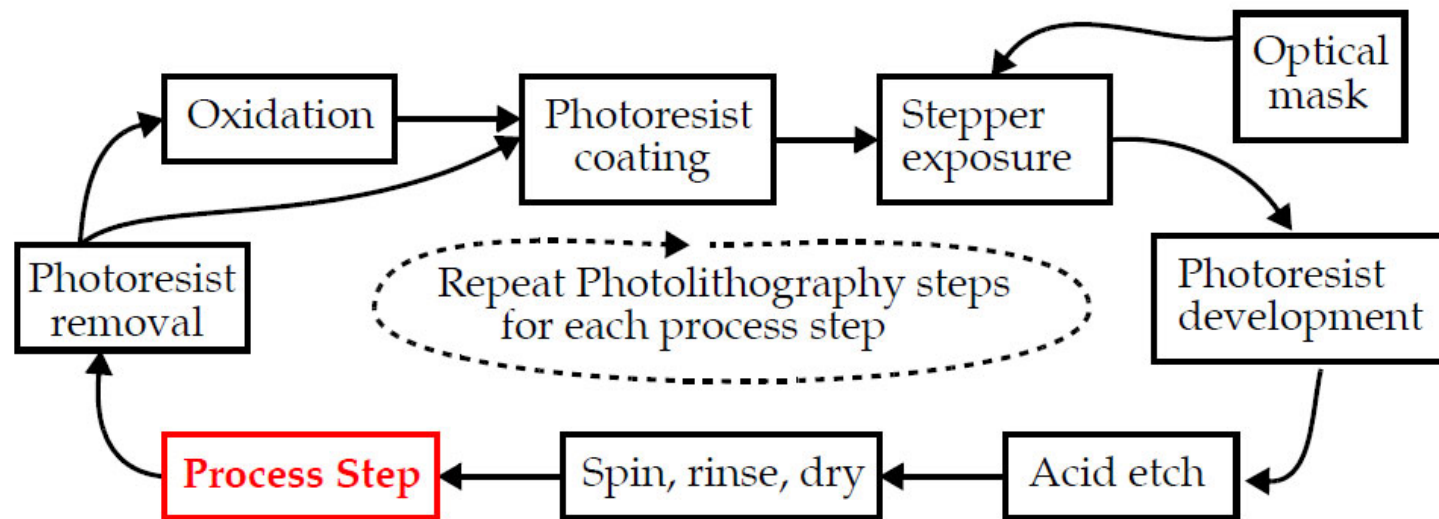


Photolithography



Ref: L.F. Thompson, An introduction to lithography, in: Introduction to Microlithography, ACS Symposium Series, Washington, DC, 1983.

Photolithography



Photolithography- Resolution

Definition of Resolution

Resolution in photolithography refers to the **smallest feature size** (minimum line width or space) that can be accurately patterned on a wafer using a given optical system and photoresist process.

It determines how **fine or dense** the circuit patterns (transistors, interconnects, etc.) can be — directly affecting the **technology node** (e.g., 180 nm, 65 nm, 5 nm, etc.).

Resolution Equation (Rayleigh Criterion)

$$R = k_1 \frac{\lambda}{NA}$$

Photolithography- Resolution

Resolution Equation (Rayleigh Criterion)

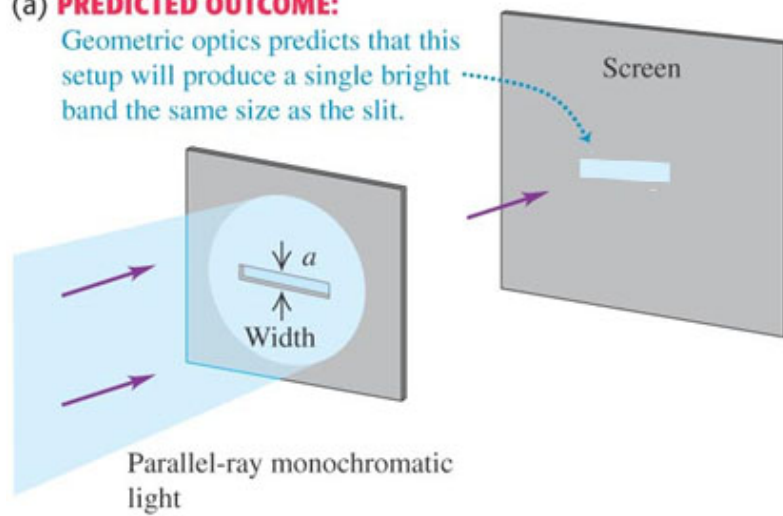
$$R = k_1 \frac{\lambda}{NA}$$

	Parameter	Description
R	Resolution	Minimum printable feature size
λ	Wavelength	Exposure light wavelength (e.g., 365 nm, 248 nm, 193 nm, 13.5 nm)
NA	Numerical Aperture	Light-gathering ability of the projection lens
k_1	Process factor	Depends on resist, illumination, and process optimization (~0.25–0.8)

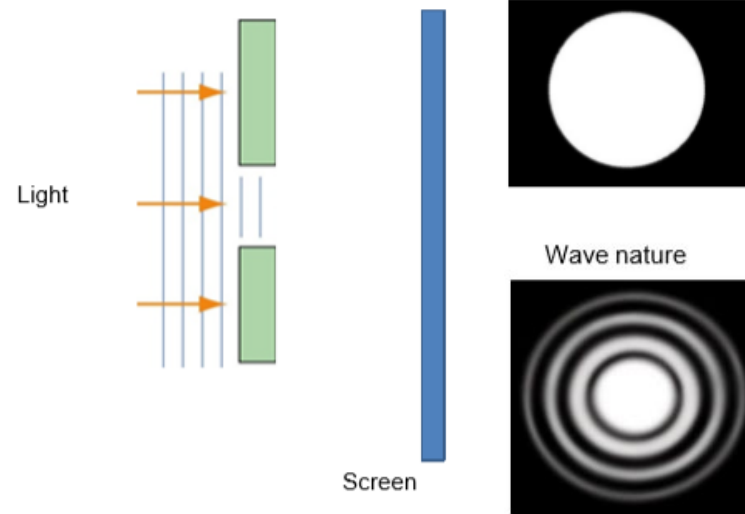
Single Slit Diffraction

(a) **PREDICTED OUTCOME:**

Geometric optics predicts that this setup will produce a single bright band the same size as the slit.



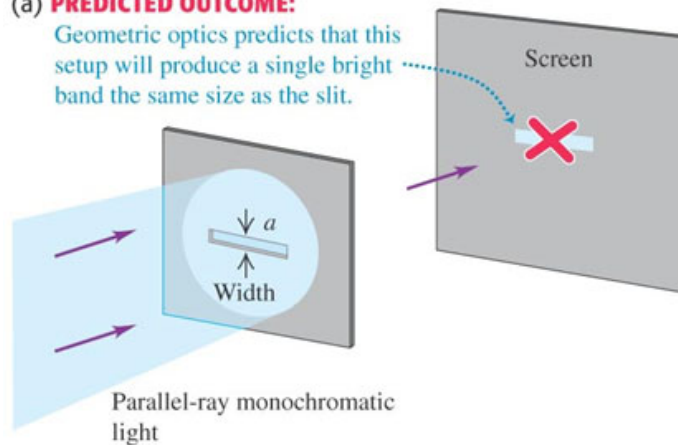
Diffraction



Single Slit Diffraction

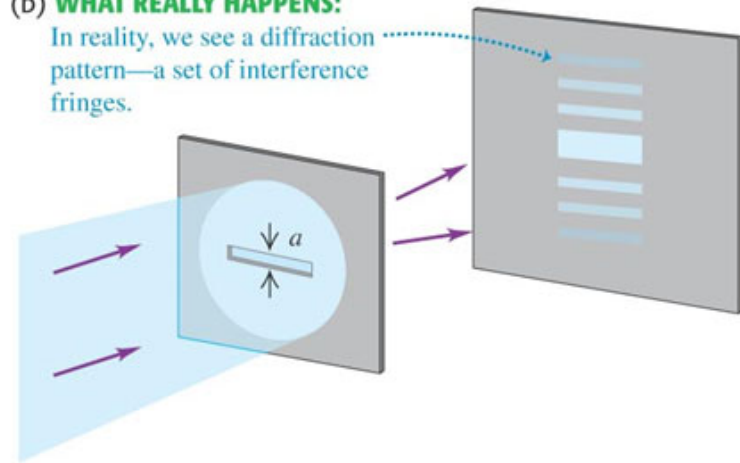
(a) **PREDICTED OUTCOME:**

Geometric optics predicts that this setup will produce a single bright band the same size as the slit.



(b) **WHAT REALLY HAPPENS:**

In reality, we see a diffraction pattern—a set of interference fringes.



According to geometric optics, a light source shining on an object in front of a screen should cast a sharp shadow. Surprisingly, this does not occur because of diffraction

Rayleigh criteria for resolution R

$$R = k_1 \frac{\lambda}{NA}$$

k_1 factor has no well-defined physical meaning.

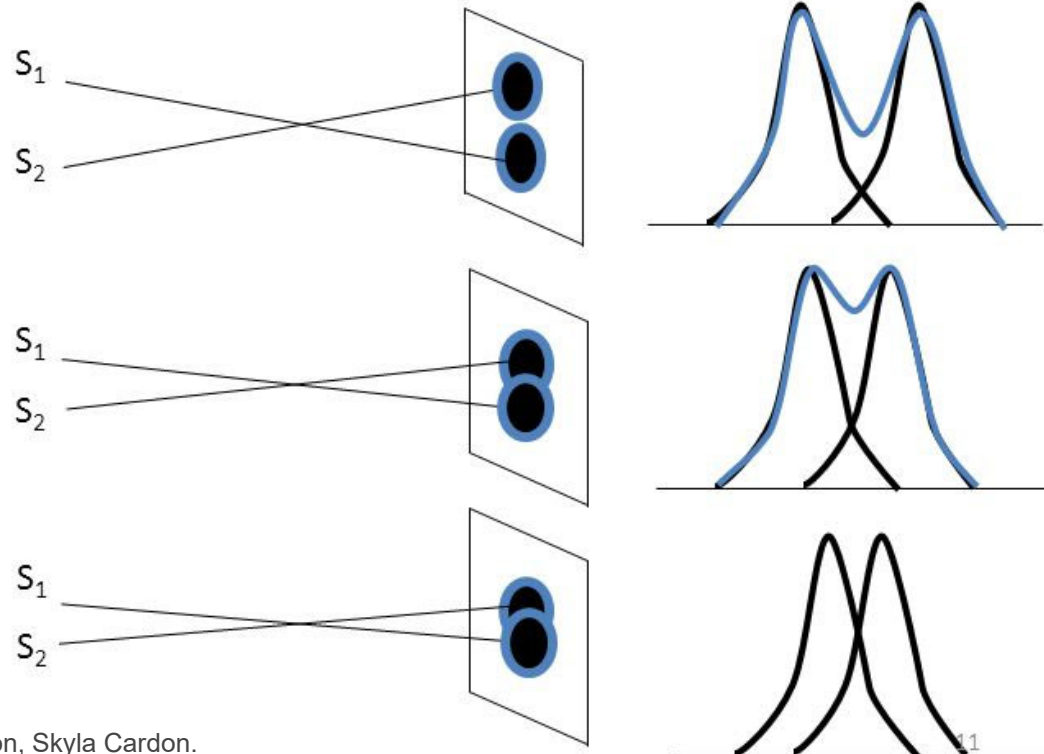
It is an experimental parameter, depends on the lithography system and resist properties.

To increase resolution,
one can:

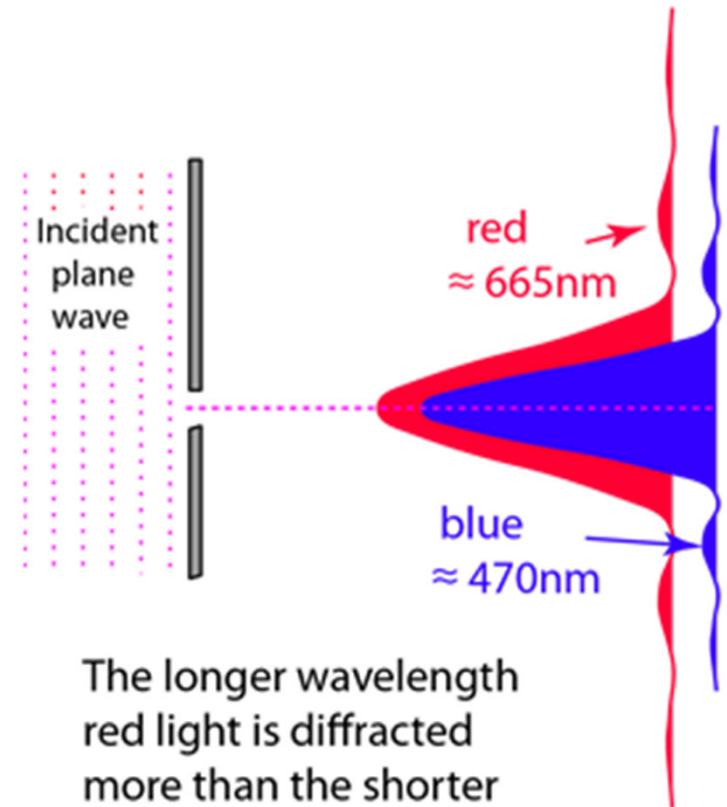
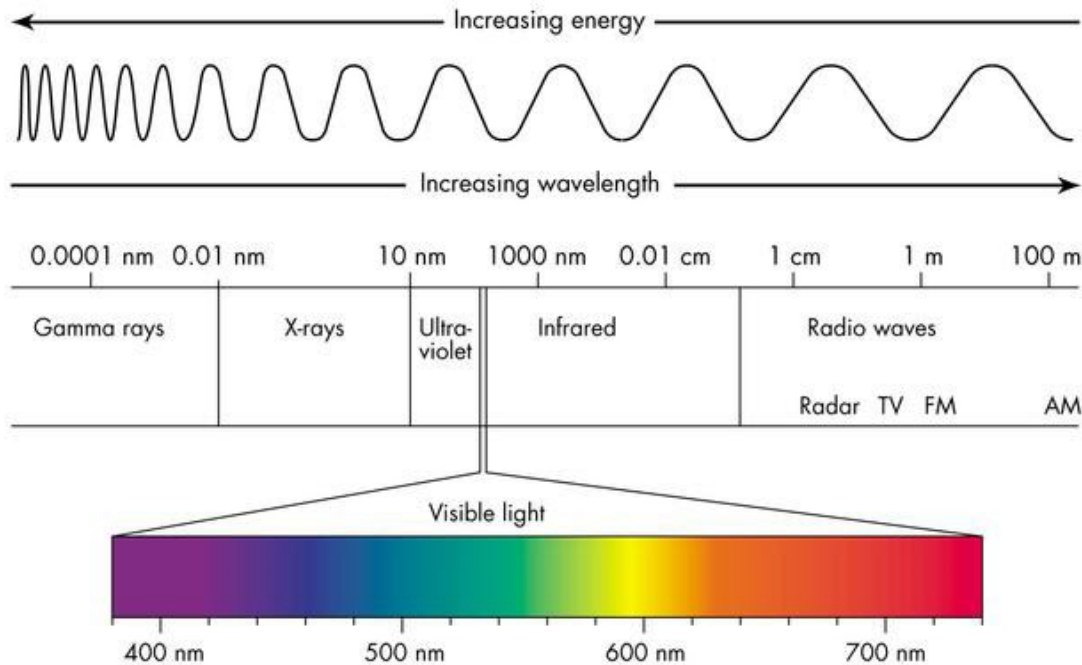
Increase NA by using large
lens and/or immersion in
a liquid ($n > 1$).

Decrease k_1 factor (many
tricks to do so).

Decrease λ (not easy,
industry still insists on
193nm).



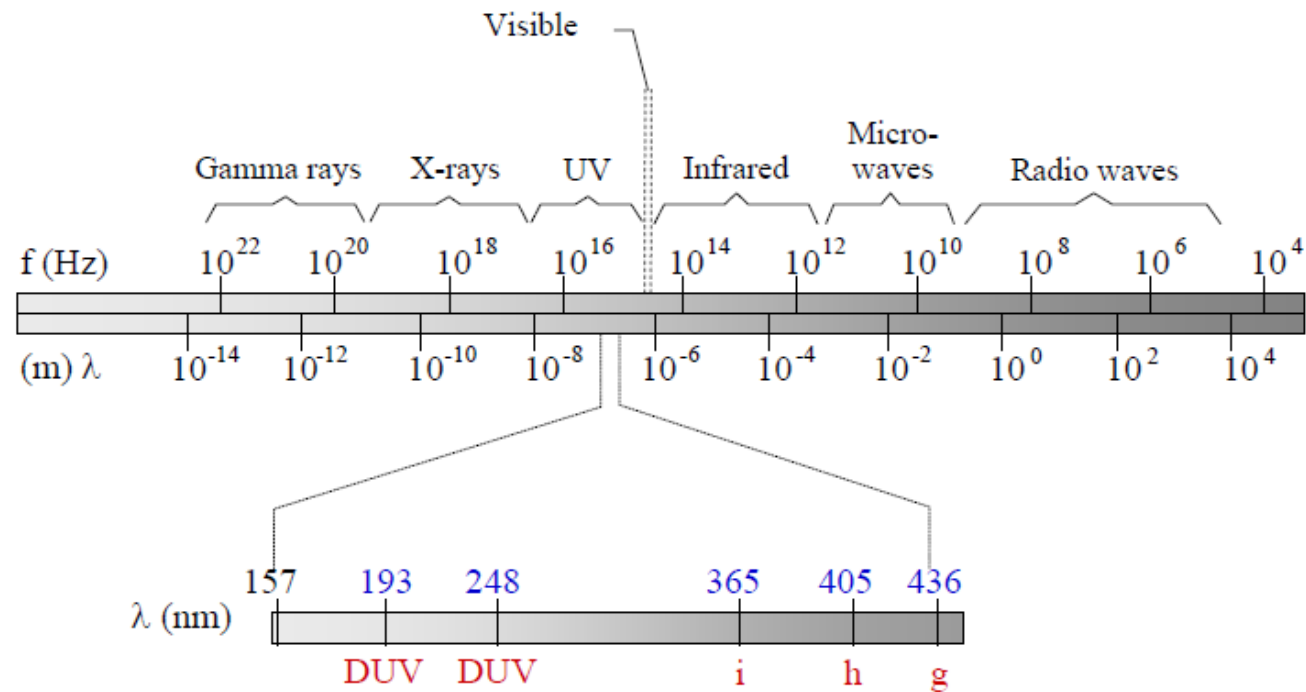
Wavelength spectrum



The longer wavelength red light is diffracted more than the shorter blue wavelengths.

Wavelength spectrum

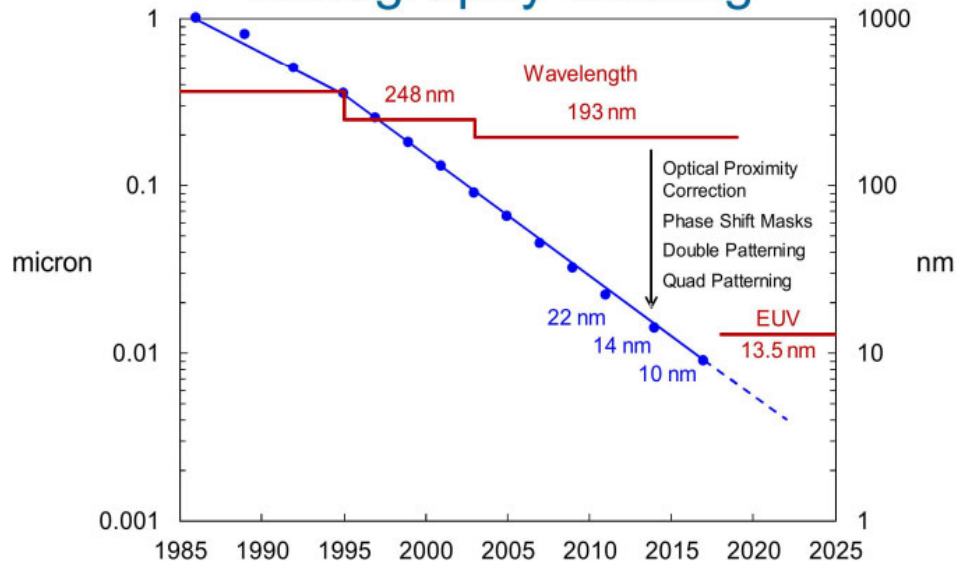
Section of the Electromagnetic Spectrum



Common UV wavelengths(10nm to 400nm) used in optical lithography.

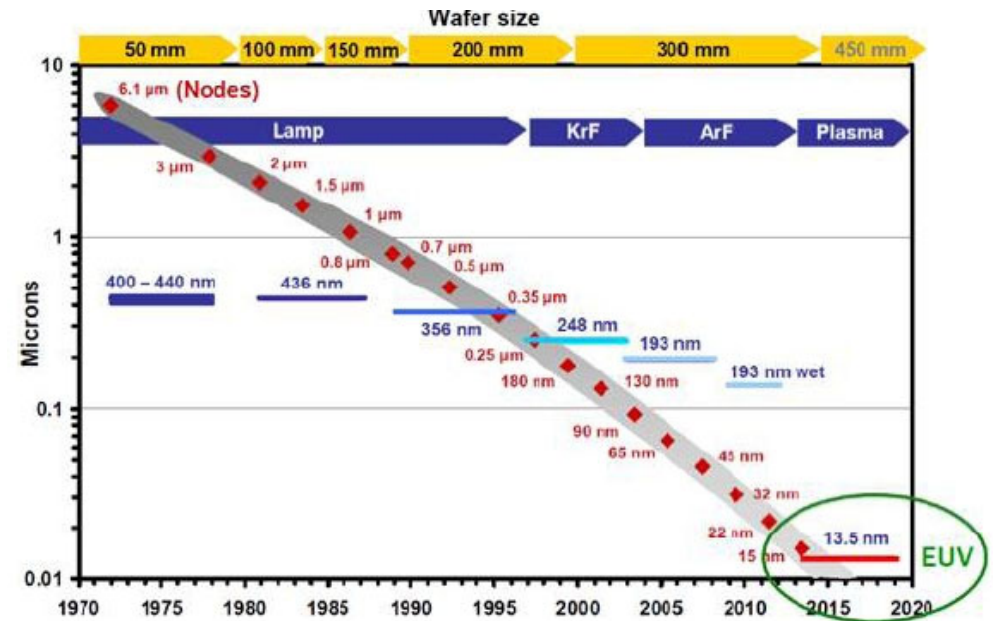
Wavelength scaling and its impacts

Lithography Scaling



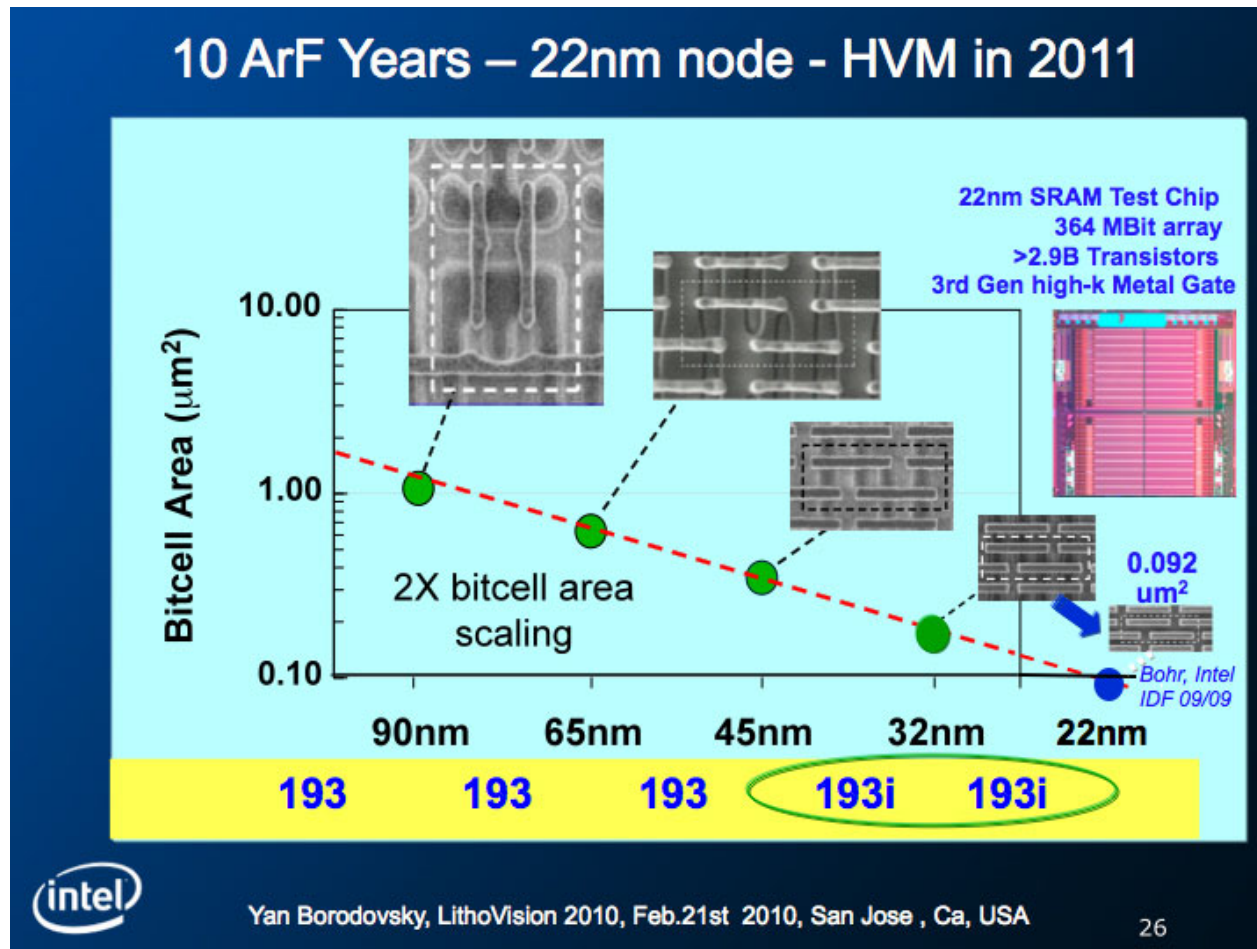
EUV will enable several more generations of scaling

$$R = k_1 \frac{\lambda}{NA}$$

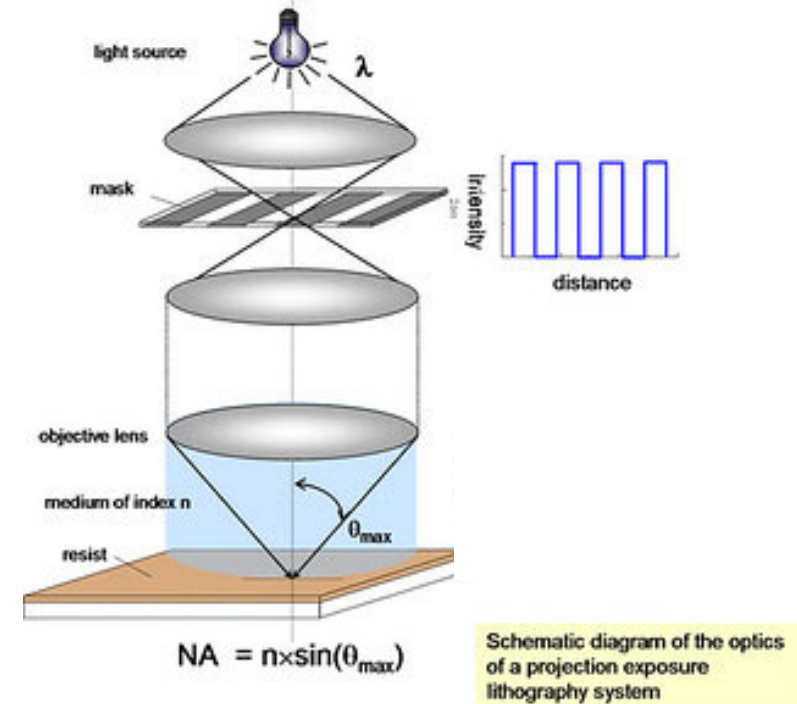
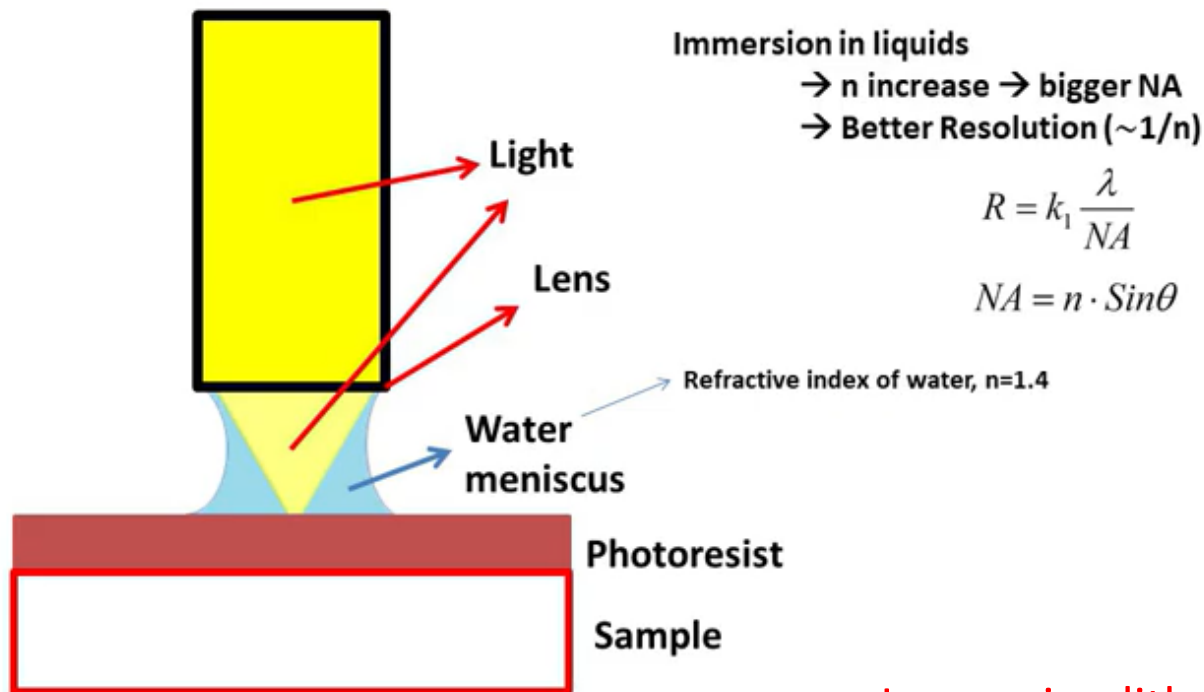


Ref: DRAM Chip market primed for long lasting resilient up cycle, 2013

193nm ArF ([Argon fluoride laser](#)) Photolithography



193nm (i) ArF Photolithography



Immersion lithography was a **key enabler** for 45 nm $\rightarrow$ 28 nm technology nodes.

193nm ArF Photolithography-Double Patterning

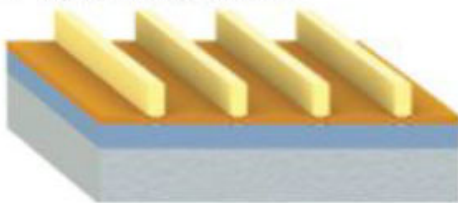
- **Litho-Etch-Litho-Etch**
- **Litho-Freeze-Litho-Etch**
- **Self-Aligned Double (Multiple) Patterning**

193nm ArF Photolithography-Double Patterning

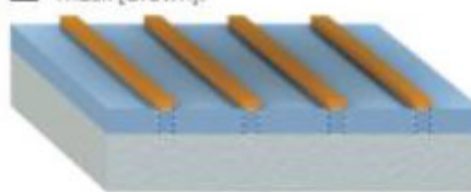
- **Litho-Etch-Litho-Etch**
 - First double patterning technique
 - Doubles the processing steps, so doubles the cost and reduces throughput
 - Requires a hard mask
 - Suffers from mask overlay issues

Litho-etch-litho-etch (LELE) is a form of double patterning. LELE is also called pitch splitting. In LELE, two separate lithography and etch steps are performed to define a single layer, thereby doubling the pattern density.

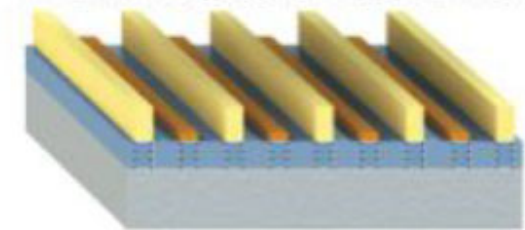
1 **Litho 1.** The first pattern [yellow] is exposed onto a hard mask.



2 **Etch 1.** The first pattern is etched into the hard mask [brown].

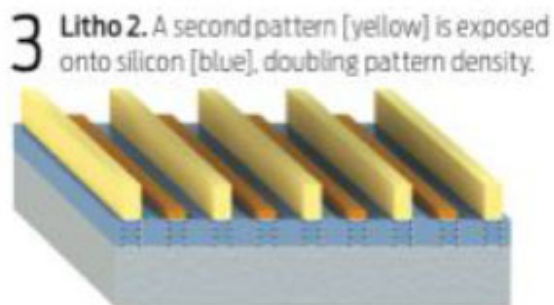
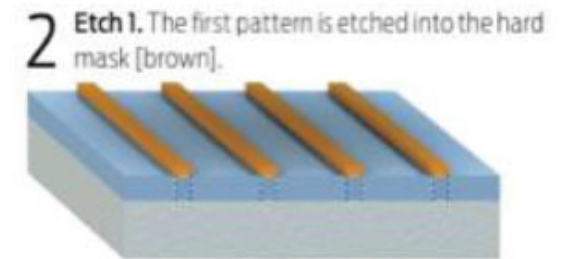


3 **Litho 2.** A second pattern [yellow] is exposed onto silicon [blue], doubling pattern density.



193nm ArF Photolithography-Double Patterning

- Litho-Etch-Litho-Etch
 - First double patterning technique
 - Doubles the processing steps, so doubles the cost and reduces throughput
 - Requires a hard mask
 - Suffers from mask overlay issues



193nm ArF Photolithography-Double Patterning

- The first double patterning technique, litho-etch litho-etch (LELE), is the most straightforward. It consists of an initial lithography step followed by an etching step and then the same process is repeated to get the final pattern.
- The negative aspects of this technique are that two masks must be used to create the final pattern. This directly increases the time and cost it takes to fabricate using double patterning because the process steps will be effectively doubled.
- Another struggle with this technique is small tolerance of the pattern overlay. Because of the small-scale patterns, a slight misalignment of the second pattern will cause a notable defect in the final pattern.

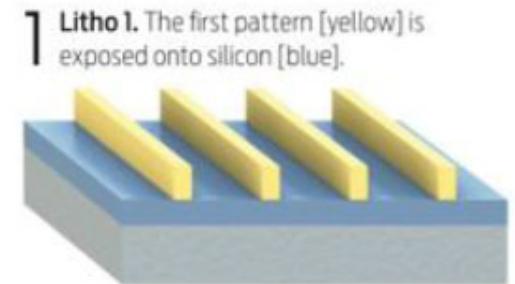
193nm ArF Photolithography-Double Patterning

Lithography → Freeze → Lithography → Etch (**LFLE**)

It is a **dual-patterning** technique developed to **reduce cost and overlay error** compared to **LELE**.

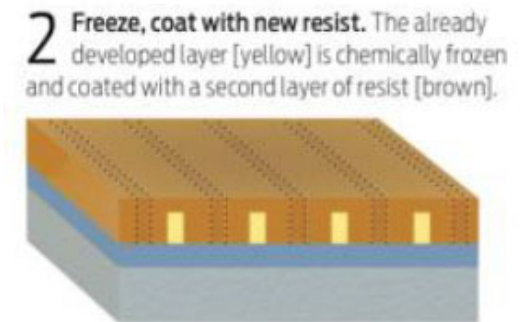
(a) First Lithography (L1)

- The first photoresist (PR1) is coated and exposed using **Mask 1**, defining the first pattern.
- The resist is **not yet developed** or etched into the substrate.



(b) Freeze

- The first resist image is “**frozen**” chemically or thermally to harden it — this makes it **insoluble** to subsequent coating and exposure steps.
- The frozen resist now acts as a **temporary hard mask**, stable enough to survive the next lithography.
- A **second photoresist (PR2)** layer is coated over the same wafer.



193nm ArF Photolithography-Double Patterning

Lithography → Freeze → Lithography → Etch (**LFLE**)

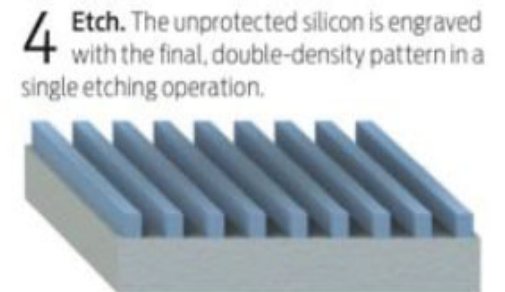
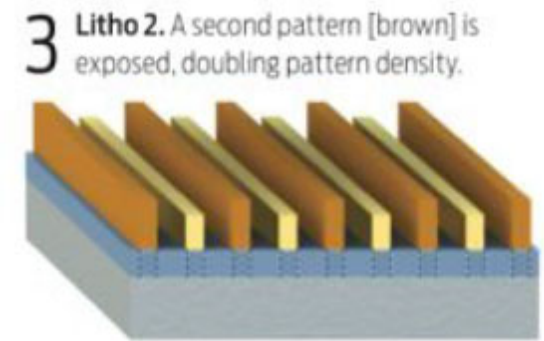
It is a **dual-patterning** technique developed to **reduce cost and overlay error** compared to **LELE**.

(c) Second Lithography (L2)

- It is exposed with **Mask 2**, defining a **complementary or shifted pattern** relative to the first one.
- PR2 is developed normally.

(d) Etch

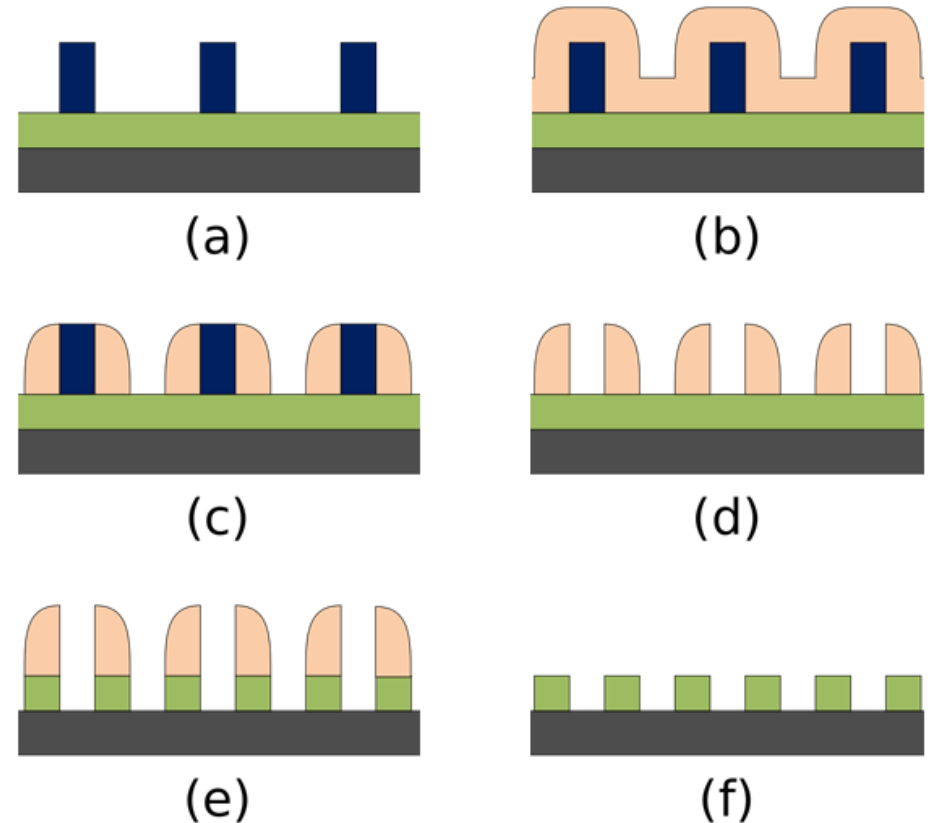
- After both patterns (frozen PR1 + developed PR2) are present, they **together define the final combined pattern**.
- A **single etch** step transfers both sets of features into the substrate.



193nm ArF Photolithography-Double Patterning

- Self-Aligned Double Patterning (SADP)

- A **key double-patterning technique** widely used in advanced CMOS fabrication (e.g., 20 nm – 10 nm nodes).
- It overcomes some of the overlay limitations of **LELE** and **LFLE** by using **sidewall spacers** instead of two separate mask alignments.



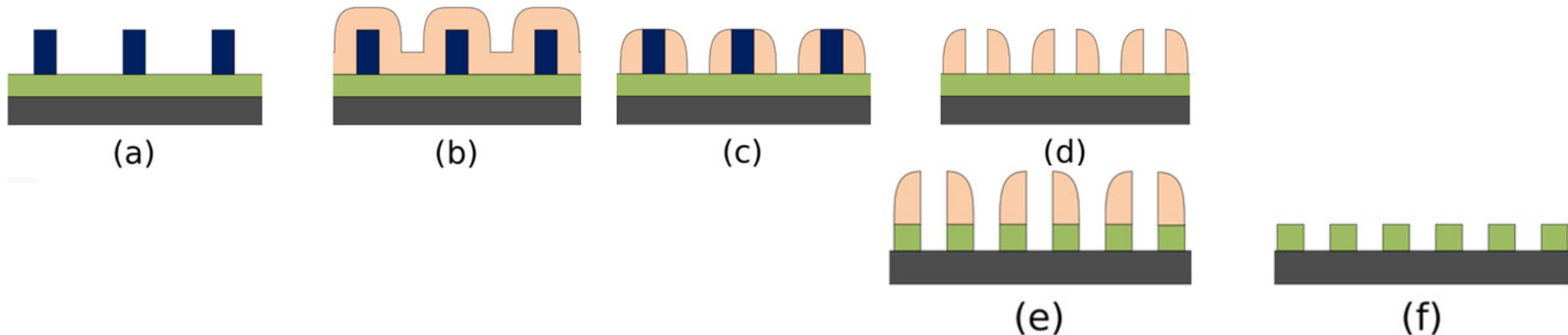
[https://commons.wikimedia.org/wiki/File:Self-aligned_\(spacer\)_double_patterning_\(SaDP\).svg](https://commons.wikimedia.org/wiki/File:Self-aligned_(spacer)_double_patterning_(SaDP).svg)

193nm ArF Photolithography-Double Patterning

• Self-Aligned Double Patterning (SADP)

Instead of using **two masks**, we use **one mask** to define “mandrel” features, then **deposit spacers** along their sidewalls, and finally **remove the mandrel**, leaving the **spacers as new, narrower features**.

These spacers are **self-aligned**, meaning they do not depend on another mask’s alignment — hence, **no overlay error** between two masks.

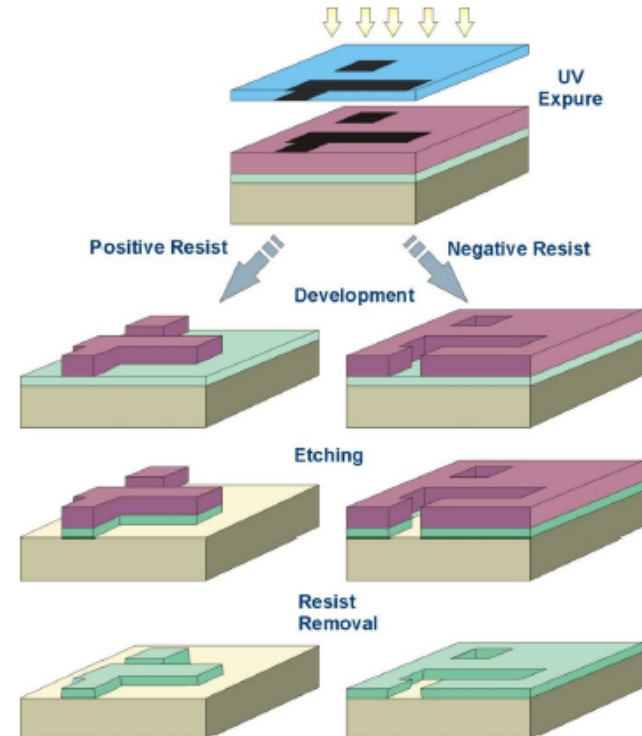


Etching (Development)

Etching is a process by which material is removed from a substrate or from thin films on the substrate surface. When a mask layer is used to protect specific regions of the wafer surface, the goal of the etching is to «precisely» remove the material which is not covered by the mask

There are two fundamental groups of etching:

- **wet etching** (liquid-based etchants)
- **dry etching** (plasma-based etchants, but not only ...)



REF:

It-fab
Italian Network for
Micro and Nano Fabrication

SUTD
SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

Etching (Development)

Etching

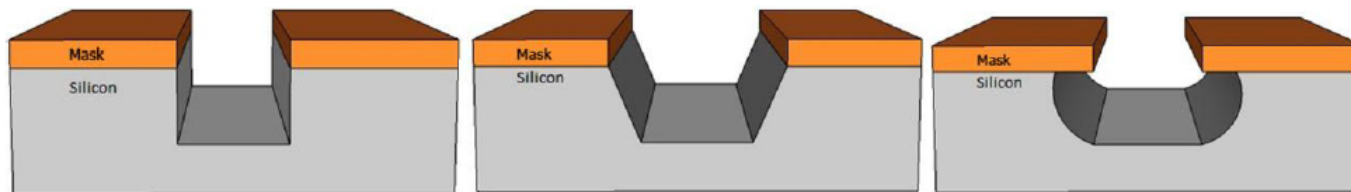
Isotropic

- etches equally in all direction
- wet etch is isotropic
- this etch leads to “undercutting”

Anisotropic

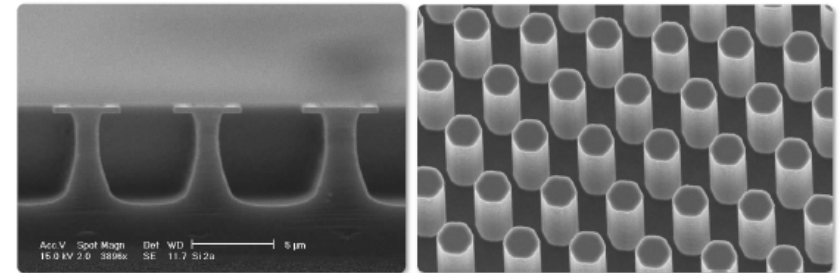
- the etch rate is dependant on the direction of the etch
- dry etch is anisotropic

Anisotropic etching has different etch rates in the lateral and vertical directions.



(a) Completely anisotropic (b) Partially anisotropic and (c) Isotropic etching of silicon

AN EXAMPLE



ISOTROPIC (LEFT) AND ANISTROPIC (RIGHT)

Etching (Development)

Etching

- Etching is the process of removing material from the substrate
- Etches can remove Si, SiO₂, polysilicon, and metal depending on what we want to accomplish

Etching

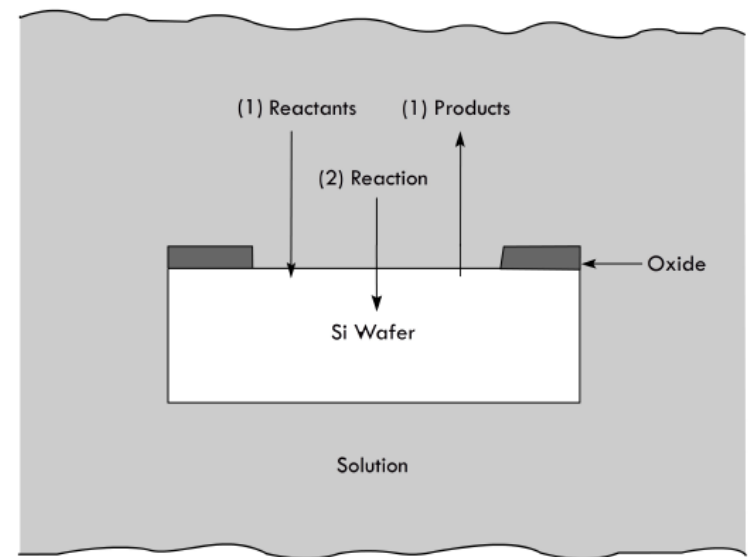
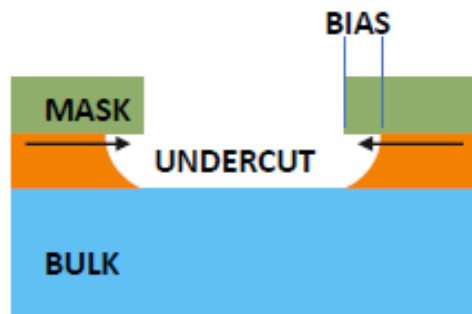
- there are two common types of etch processes

“Wet Etch” - also called Chemical Etch

- this uses Hydrofluoric Acid (HF acid)
- the wafer is submerged in the acid
- simple, but produces toxic waste

Etching (Development)-Wet

1. Wet etching is used for large pattern sizes, usually larger than $2\ \mu\text{m}$.
2. It is an isotropic process - sloped sidewalls rather than straight walls.
3. Wet etch has to be combined with subsequent rinse and dry steps. This increases chances of defects or contamination.
4. Hazardous chemicals and conditions are used, so safety is an issue. Safe disposal of chemicals is essential.
5. Undercutting and resist peel off can happen if time is not controlled or etch conditions change during process.



Schematic of the wet etching process.

Etching (Development)

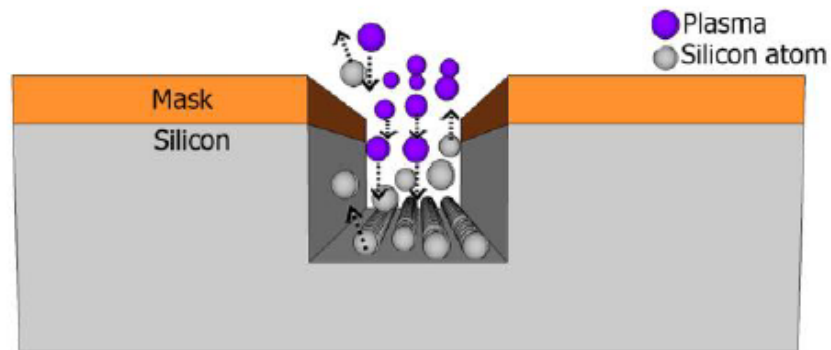
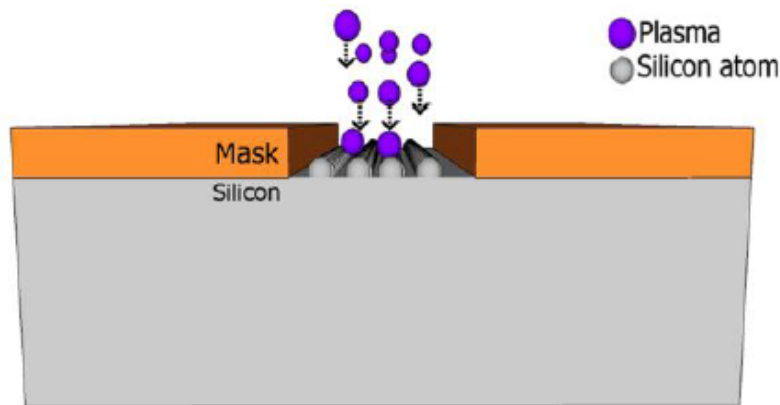
“ Dry Etch”

- Reactive-Ion Etch (RIE)
 - the chemical reaction forms a new compound that is loose and may be removed
 - since the ions move up and down, we can make a very vertical etch pattern
- Plasma Etch
 - plasma is a charged gas which has excited ions
 - plasma can be *moved* by applying an E-field

Etching (Development)-Dry

Physical dry etching:

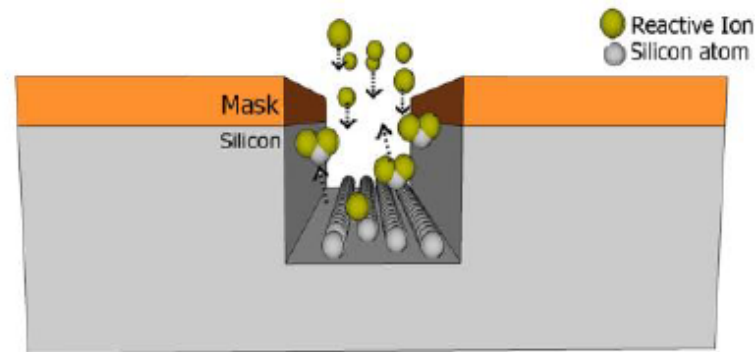
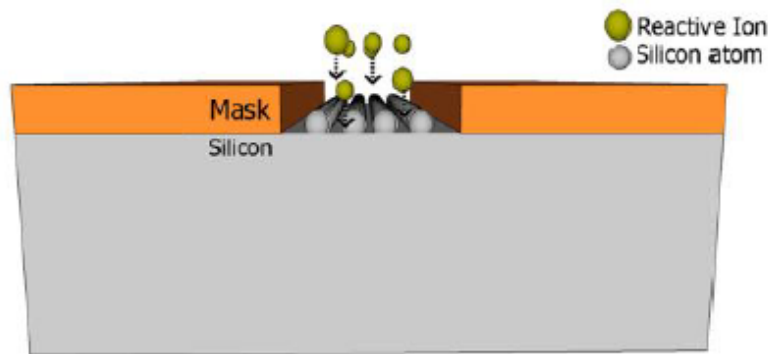
Physical dry etching requires high energy kinetic energy (ion, electron, or photon) beams to etch off the substrate atoms. When the high energy particles knock out the atoms from the substrate surface, the material evaporates after leaving the substrate. There is no chemical reaction taking place and therefore only the material that is unmasked will be removed. The physical reaction taking place is illustrated in Figure.



Etching (Development)-Dry

Chemical dry etching:

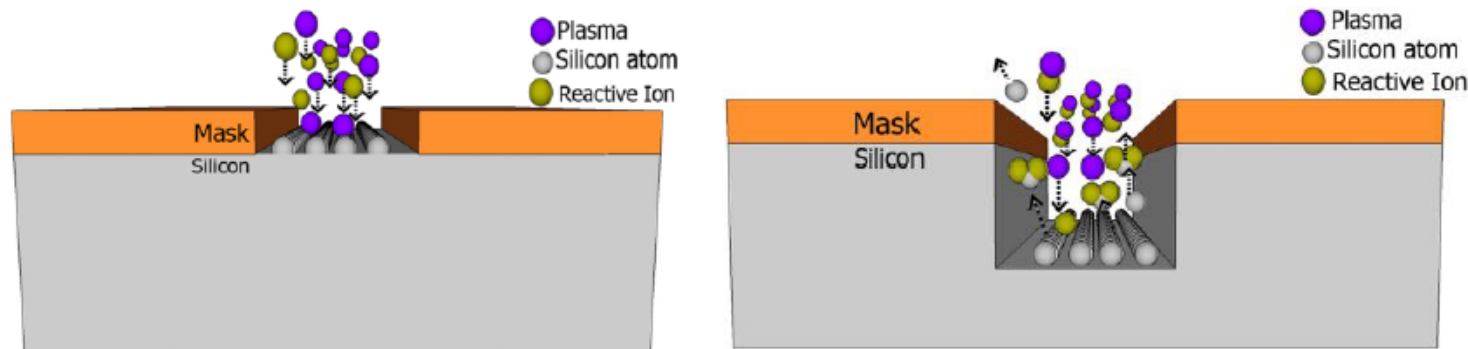
Chemical dry etching (also called vapor phase etching) does not use liquid chemicals or etchants. This process involves a chemical reaction between etchant gases to attack the silicon surface. The chemical dry etching process is usually isotropic and exhibits high selectivity. Anisotropic dry etching has the ability to etch with finer resolution and higher aspect ratio than isotropic etching. Due to the directional nature of dry etching, undercutting can be avoided.



Etching (Development)-Dry

Reactive Ion Etching:

Reactive ion etching (RIE) uses both physical and chemical mechanisms to achieve high levels of resolution. The process is one of the most diverse and most widely used processes in industry and research. Since the process combines both physical and chemical interactions, the process is much faster. The high energy collision from the ionization helps to dissociate the etchant molecules into more reactive species.



The RIE process. This process involves both physical and chemical reactions to etch off the silicon.



Plasma

- Plasma is a fundamental state of matter—sometimes called the **fourth state of matter**—alongside solid, liquid, and gas.
- Plasma is an **ionized gas** consisting of:
 - Free electrons, Positive ions,**
Neutral atoms or molecules (in lesser proportion)
- It contains charged particles, plasma conducts electricity, responds to magnetic/electric fields, and emits light.

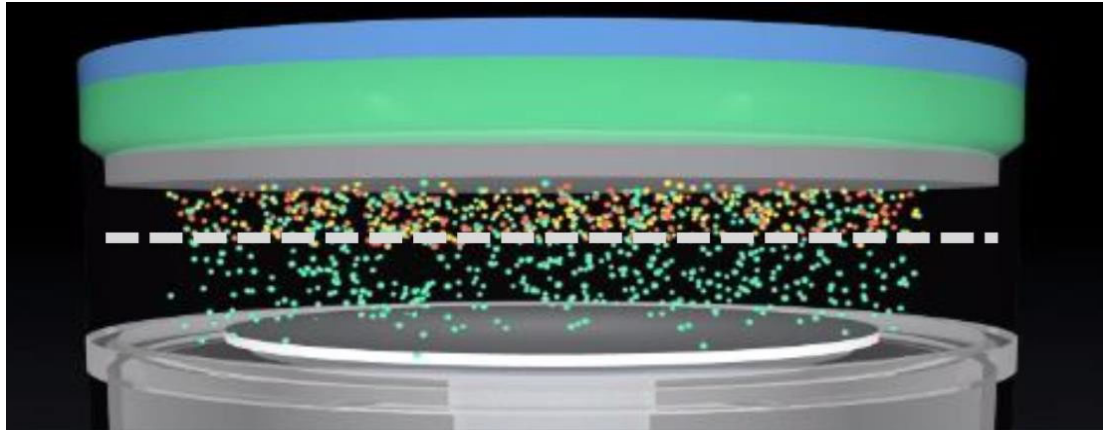
Plasma – Generation

- When a gas is supplied with enough energy (heat, strong electric field, or electromagnetic radiation), its atoms/molecules lose electrons → creating a mixture of ions and free electrons.

Plasma – Enhanced Processes in Microelectronics

- Etching
 - dry etching
 - reactive ion etching (RIE)

Plasma Etching



(Source: Applied Materials)

Plasma etching is a **dry etching process** that uses plasma (ionized gas) to **remove material from a wafer surface** in a controlled, patterned way.

The **combination of chemical reaction and physical ion bombardment** removes the material from exposed areas of the wafer.

Wet Etch and Dry Etch

Feature	Wet Etching	Dry Etching
Medium	Liquid chemicals	Plasma (gas phase)
Etch Rate	Fast (100 – 10,000 nm/min)	Slower (10 – 1000 nm/min)
Directionality	Isotropic (poor control)	Anisotropic (good control)
Control	Moderate (affected by diffusion)	Excellent (nm-scale precision)
Damage Risk	Minimal chemical damage	Plasma damage possible
Applications	Bulk removal, MEMS, simple layers	IC nanometer-scale patterning

Etchants Applications



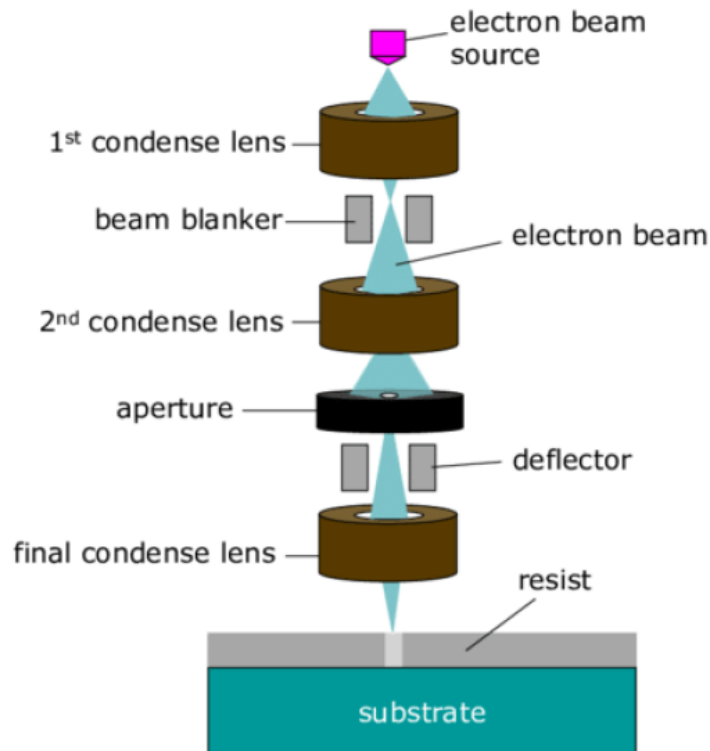
- Potassium hydroxide (KOH) Used to etch bulk Silicon
- Hydrofluoric Acid (HF) Used to etch Silicon Dioxide and Tungsten
- Sulfur Hexafluoride (SF₆) Used to etch Silicon Nitride Used
- Boron Trichloride (BCl₃) to etch Aluminum

E-beam writing/lithography

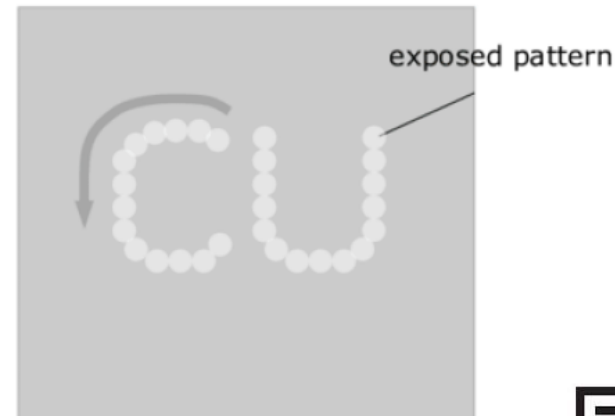
- **Electron-beam lithography** (often abbreviated as **e-beam lithography** or **EBL**) is the practice of scanning a focused beam of electrons to draw custom shapes on a surface covered with an electron-sensitive film called a resist (exposing).
- The electron beam changes the solubility of the resist, enabling selective removal of either the exposed or non-exposed regions of the resist by immersing it in a solvent (developing). The purpose, as with [photolithography](#), is to create very small structures in the resist that can subsequently be transferred to the substrate material, often by etching

E-beam writing/lithography

It's a **maskless, high-resolution** lithography technique used mainly for **research, prototype fabrication, and nanostructure patterning** in semiconductor and nanotechnology applications.

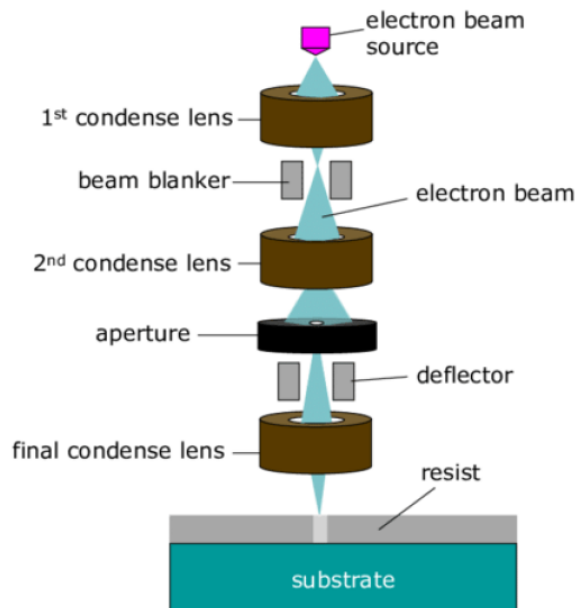


Can use magnetic or electric fields for lenses

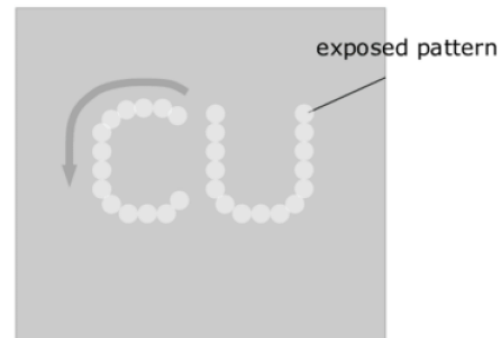


E-beam writing/lithography

- **Electron Beam Lithography (EBL)** uses a **focused beam of high-energy electrons** to directly write patterns onto an **electron-sensitive resist** (usually PMMA or HSQ) on the wafer surface — **without using a photomask**. PMMA, or polymethyl methacrylate is highly transparent and is an optical polymer, making it suitable for applications requiring clear visibility.
- Because the electron beam can be **narrowly focused (few nanometers)**, EBL achieves **extremely fine pattern resolution** — much smaller than optical lithography limits.

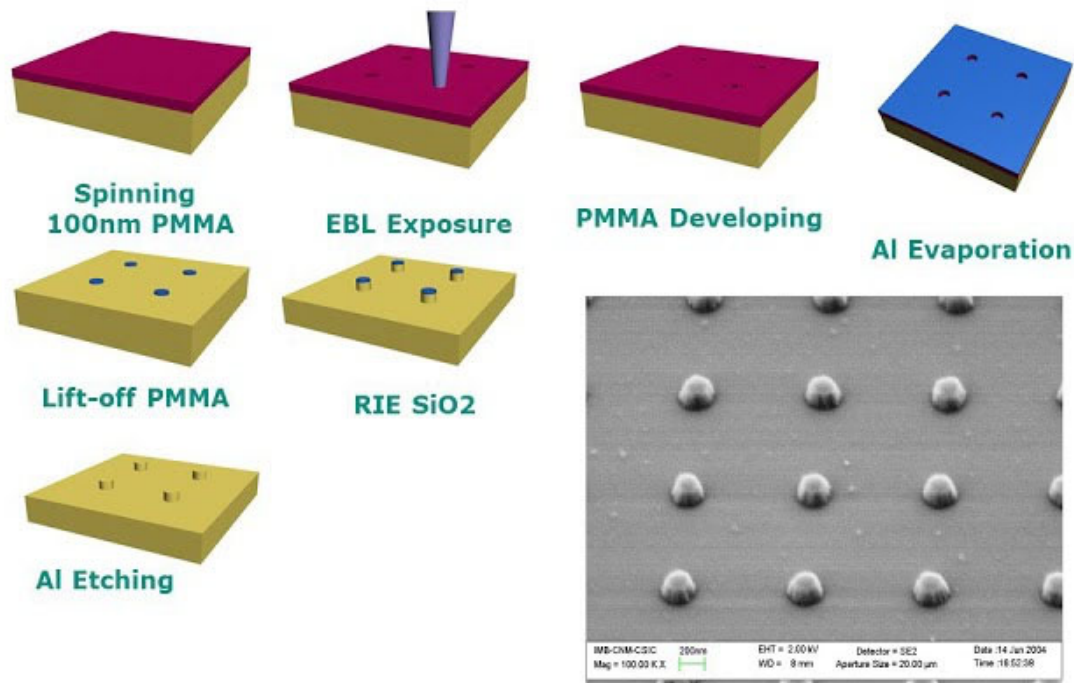


Can use magnetic or electric fields for lenses

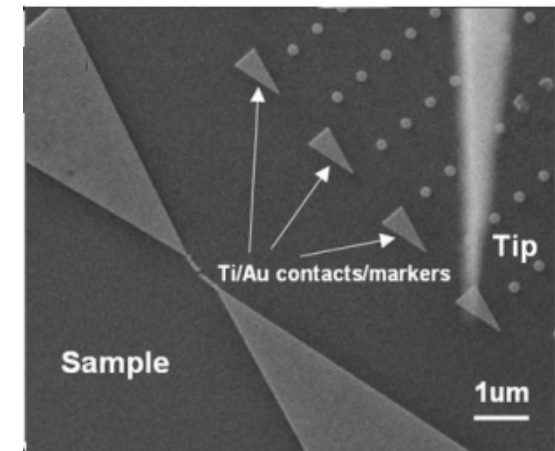


E-beam writing/lithography

EBL: Complete process



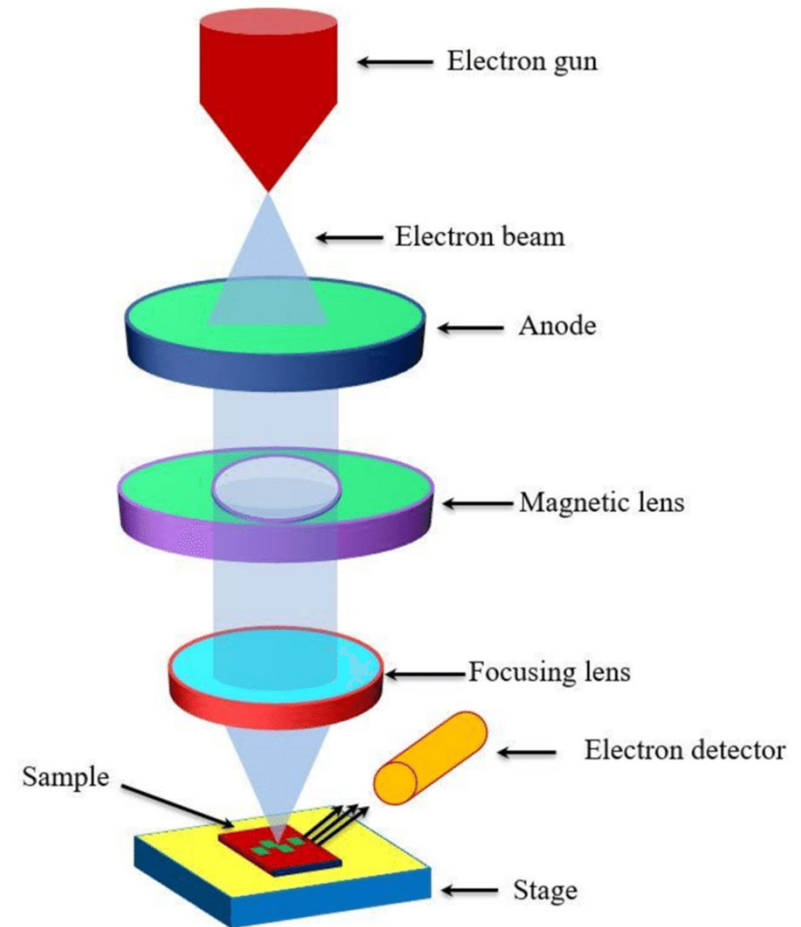
A **focused electron beam** scans the surface following a **computer-controlled pattern**. Exposed regions undergo **chemical changes**.



F. Torres, UA De Barcelona-Electron beam lithography

E-beam writing/lithography

- E-beam resist **Polymethyl methacrylate** (PMMA).
- Spin coating
- No mask required
- Slow process
- High resolution
- Suitable for custom based pattern
- Electron scattering is the main limitation of writing small features.



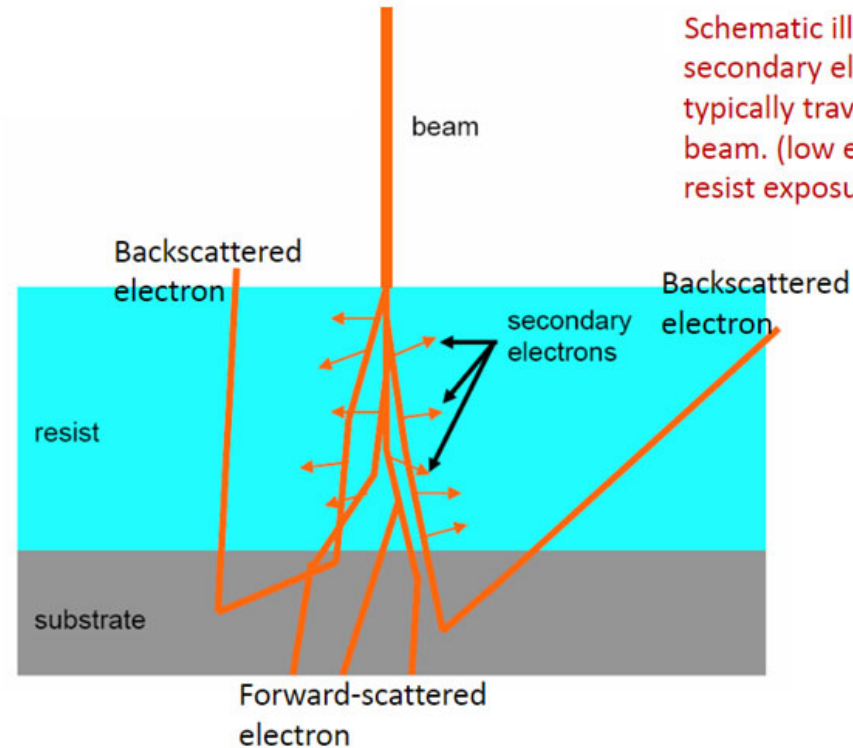
Pijush MS thesis, 2016

https://www.researchgate.net/figure/Schematic-diagram-of-the-E-beam-lithography_fig3_322317317

E-beam writing/lithography

Resolution limits from electron scattering:

- Forward scattering (in resist)
- Backscattering (from substrate)



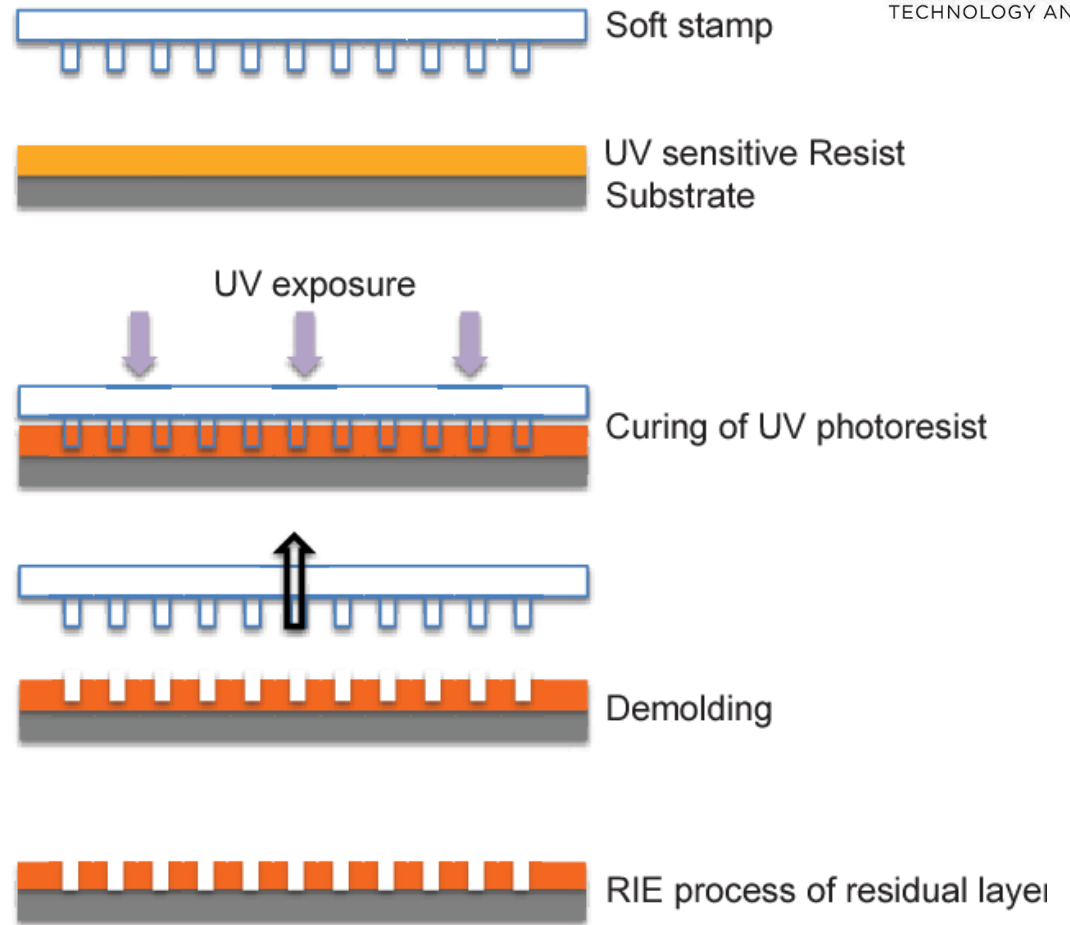
Schematic illustration of secondary electrons (SE), which typically travel/diffuse normal to the beam. (low energy SE is responsible for resist exposure)

Whether it is forward scattering or beam spot size that limits the minimum line-width, the effect of SE diffusion is to further broaden the line/reduce the resolution.

https://studylib.net/doc/9599457/electron-beam-lithography_3

Nanoimprinting

- Nanoimprint lithography (NIL) is a **method of fabricating nanometer scale patterns**.
- It is a simple nanolithography process with low cost, high throughput and high resolution.
- It creates patterns by mechanical deformation of imprint resist and subsequent processes.



Soft UV Nanoimprint Lithography: A Tool to Design
Plasmonic Nanobiosensors-G. Barbillon, 2011